



LMPS SPECIFICATION

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LMPS API (v 8.6)

1. Draft Change log

1.1. draft 8.3

- Add example message
- Add Max-length for data field
- add Tag 54 to Static QR code structure
- add CRC Implement
- merge Static and dynamic QR code to only QR code structure

1.2. draft 8.4

- extend member_code(frommember, tomember) max-length to 16

1.3. V 8.5

- Add field “fromuserfullname” to request message for Cross-border transaction
- Add inquiry step on [16. Thailand cross border payment](#)

1.4. V8.5.3

- Add lookup and payment flow and new variable for CrossBorder to 16. Thailand cross border payment
- Remote tomember
- Add Finlink to Settlement Report

1.5. V8.5.6

- Add [China cross border payment](#)

1.6. V8.6

- Update Lao QR code Inquiry Project
- Update Lao QR payment confirmation process
- Update QR code structure

1.7. V8.6.2

- Update Crosborder Response
- Add 8. Currency reimbursement report

1.8. V8.6.3

- Update Crosborder When requested by UnionPay "frommember": "CHINA_UPI"

1.9. V8.6.4

- Add [Cross border API for Settlement Bank](#)

1.10. V8.6.5

- Add [Cambodia cross border payment](#)
- Add [Exchange THB API for Service Bank](#)
- Remove [Cross border API for Settlement Bank](#)

1.11. V8.6.6

- Add [Incoming Inquiry and Transfer \(UnionPay-Lao\)](#)

1.12. V8.6.7

- Add [Example Response coming from Cambodia Outgoing inquiry](#)

2. Member's Tasks

- Process fee:
 - a) Calculate and display fee according to "feelist" received when inquiry.
 - b) Debit customer's account accordingly.
 - c) Submit "fee" in a transfer request message to double check with LAPNET.
- Process failed transfers (more details in "8. How to proceed when transfer failed"):
 - a) Source member check field "shouldrevert" and process accordingly.
 - b) Source member creates a system to periodically check failed transactions having "shouldrevert" is false.
 - c) Source member changes mobile banking error message according to "shouldrevert"
 - d) Destination member provides "shouldrevert" when transfer failed.
 - e) Destination member creates a system to periodically check its own destination account and update "result" and "shouldrevert" accordingly.
- Implement "6. Check Transaction Status"
- Change "purposerequired" data type from Boolean to Integer

3. Design Considerations

- Message spec must be simple and easy to implement, leading to shorter development time
- Using public/private keys to generate a message signature instead of symmetric 3DES/AES prevents anyone from faking the message, not even LAPNET can fake members' messages because only members hold their own private keys. This is important especially when performing a reconciliation.
- Because message signature is secure enough, session key exchange will not be needed, reduce number of messages and the system complexity
- LMPS server supports HTTP Keep-Alive to reduce connect/disconnect overhead
- To make the implementation process easier, destination members do not have to verify the source member's signature, only verify LAPNET signature.

4. Member Registration

4.1. Provide the public key

The example below generates a key pair using software. For a production key pair, using HSM is highly recommended.

1. Generate RSA public-private key pair, 2048 bits. E.g.:
`$ openssl genrsa -des3 -out keypair.pem 2048`
2. Export public key in PEM format. E.g.:
`$ openssl rsa -in keypair.pem -outform PEM -pubout -out public.pem`
3. Optionally, export private key in PEM format to use with Java:
`$ openssl pkcs8 -topk8 -inform PEM -outform pem -in keypair.pem -out private.key -nocrypt`
4. Submit only the public key to LAPNET.

4.2. Provide URL for LAPNET to send requests to

LAPNET will send requests via HTTP, members need run a web server and provide URL for:

1. Incoming Inquiry. E.g., **`http://<MEMBER-IP>:8095/inquiry`**
2. Incoming transfer/payment. E.g., **`http://<MEMBER-IP>:8095/transfer`**
3. Incoming transaction notification. E.g., **`http://<MEMBER-IP>:8095/notify`**

5. Commands

5.1. Outgoing Inquiry

- When members want to transfer or make a payment, use this command to check for a destination account.
- Members need to build a message body and sign using the private key, then put in the header (X-Source-Request-Signature) and send it to LAPNET.
- After receiving a response from LAPNET, members need to verify LAPNET signature in the header (X-LAPNET-Request-Signature) of the body using LAPNET's public key.
- Members are recommended to store all signatures for reconcile purpose, signatures prove that messages really coming from LAPNET

5.1.1. Request to send to LAPNET

5.1.1.1. URL

- Development: **http://<LAPNET-IP>:8055/inquiry**
- Production: **http://<LAPNET-IP>:8055/inquiry**

5.1.1.2. Header

Content-Type: application/json

X-Source-Request-Signature: <<source bank signature>>

5.1.1.3. Body

Field	Type	Max-Length	Description
frommember	String	16	E.g. BCEL, LDB, APB, LVI
fromuser	String	64	E.g. adsavin@gmail.com
fromuserfullname	String(English)	256	E.g. ADSAVIN THEPPHAKAN MR
fromaccount	String	32	E.g. 010120000123456001
totype	String	16	ACCOUNT/CARD/QR
toaccount	String	2048	Account number or card number or full QR data
tomember	String	16	Only required if totype is ACCOUNT
reference	String	64	Member-generated unique reference
time	String	24	e.g. 2020-10-09 05:34:11

5.1.1.4. Example

```
{
  "frommember": "BCEL",
  "fromuser": "2022456699",
  "fromuserfullname": "ADSAVIN THEPPHAKAN MR",
  "fromaccount": "010120000005477001",
  "tomember": "LVI",
  "toaccount": "1234567890",
  "totype": "ACCOUNT",
  "reference": "ABCDEFGHJKLMN",
  "time": "2020-10-09 05:34:11"
}
```

5.1.2. Response coming from LAPNET

5.1.2.1. Header

Content-Type: application/json

X-Source-Request-Signature: <<original request signature>>

X-LAPNET-Request-Signature: <<LAPNET request signature>>

X-Destination-Response-Signature: <<destination response signature>>

X-LAPNET-Response-Signature: <<LAPNET response signature>>

X-LAPNET-Transaction-ID: <<Sequence transaction ID generate by LAPNET>>

5.1.2.2. Body

Field	Type	Max-Length	Description
result	String	64	See Result Table
reference	String	64	Original request reference
time	String	25	Response time
originalmessage	JSON object		Original request message
accountname	String	128	Account name, card name, merchant name
accountlocation	String (Optional)	256	
accountphone	String (Optional)	24	
invoice	String (Optional)	128	
terminal	String (Optional)	128	
amount	Number (24,2)		If transaction has fixed amount, set it here
purposerequired	Boolean (Optional)	-	- For type ACCOUNT and CARD, the default value is true. - For type QR, default value is false
purpose	String (Optional)	512	If transaction has a fixed purpose, set it here
accountccy	String separated by comma	16	E.g. "LAK", or "LAK,USD"
feelist	JSON {CCY: [{from, feeamount, feepercent}, ...]}		Fee list injected by LAPNET. E.g. <pre> "feelist": { "LAK": [{"from": 0, "feeamount": 1000}, {"from": 2000001, "feeamount": 1500}, {"from": 3000001, "feeamount": 2500},] }</pre>

5.1.2.3. Example

```

{
  "result": "OK",
  "reference": 'ABCDEFGHijklmn',
  "time": "2020-10-30 11:12:15",
  "originalmessage": {
    "frommember": "BCEL",
    "fromuser": "2022456699",
    "fromuserfullname": "ADSAVIN THEPPHAKAN MR",
    "fromaccount": "010120000005477001",
    "tomember": "LVI",
    "toaccount": "1234567890",
    "totype": "ACCOUNT",
    "reference": "ABCDEFGHijklmn",
    "time": "2020-10-09 05:34:11"
  },
  "accountname": "ADSAVIN THEPPHAKAN",
  "accountccy": "LAK,USD",
  "feelist": {
    "LAK": [
      {"from": 0, "feeamount": 1000},
      {"from": 2000001, "feeamount": 1500},
      {"from": 3000001, "feeamount": 2500},
    ]
  }
}
```



```

    ]
  }
}

```

5.2. Incoming Inquiry

- When someone wants to transfer money to members, LAPNET will send a request to members' web servers.
- When receiving a request, members must check the LAPNET signature to confirm that it is a legitimate request.
- In case something went wrong, e.g. cannot contact a source member, or an incorrect signature, LAPNET will send transaction status by calling member's notification URL.

5.2.1. Request coming from LAPNET

5.2.1.1. URL

Members will need to provide the URL to LAPNET

5.2.1.2. Header

Content-Type: application/json

X-Source-Request-Signature: <<source bank signature>>

X-LAPNET-Request-Signature: <<LAPNET request signature>>

X-LAPNET-Transaction-ID: <<Sequence transaction ID generate by LAPNET>>

5.2.1.3. Body

Field	Type	Max-Length	Description
frommember	String	16	E.g. BCEL, LDB, APB, LVI
fromuser	String	64	E.g. narin@gmail.com
fromuserfullname	String(English)	256	E.g. ADSAVIN THEPPHAKAN MR
fromaccount	String	32	E.g. 010120000123456001
totype	String	16	ACCOUNT/CARD/QR
toaccount	String	2048	Account number or card number or full QR data
tomember	String	716	Only required if totype is ACCOUNT
reference	String	64	Member-generated unique reference
time	String	25	e.g: 2020-10-09 05:34:11

5.2.1.4. Example

```

{
  "frommember": "BCEL",
  "fromuser": "2022456699",
  "fromuserfullname": "ADSAVIN THEPPHAKAN MR",
  "fromaccount": "010120000005477001",
  "tomember": "LVI",
  "toaccount": "1234567890",
  "totype": "ACCOUNT",
  "reference": "ABCDEFGHIIJKLMN",
  "time": "2020-10-09 05:34:11"
}

```

5.2.2. Response to send to LAPNET

Destination bank will verify LAPNET request signature (X-LAPNET-Request-Signature) and process the request then response to LAPNET with its signature (X-Destination-Response-Signature), LAPNET will verify X-Destination-Response-Signature then forward with header X-LAPNET-Response-Signature

5.2.2.1. Header

Content-Type: application/json

X-Source-Request-Signature: <<original request signature>>

X-LAPNET-Request-Signature: <<LAPNET request signature>>

X-Destination-Response-Signature: <<destination response signature>>

5.2.2.2. Body

Field	Type	Max-Length	Description
result	String	64	See Result Table
reference	String	64	Original request reference
time	String	25	Response time
originalmessage	JSON object		Original request message
accountname	String	128	Account name, card name, merchant name
accountlocation	String (Optional)	256	
accountphone	String (Optional)	24	
invoice	String (Optional)	128	
terminal	String (Optional)	128	
amount	Number (24,2)		If transaction has fixed amount, set it here
purposerequired	Integer 0 or 1 (Optional)	-	- For type ACCOUNT and CARD, default value is 1. - For type QR, default value is 0 Use Integer instead of Boolean to make database record easier
purpose	String (Optional)	512	If transaction has a fixed purpose, set it here
accountccy	String separated by comma	16	E.g. "LAK", or "LAK,USD"

5.2.2.3. Example

```
{
  "result": "OK",
  "reference": 'ABCDEFGHijklmn',
  "time": "2020-10-30 11:12:15",
  "originalmessage": {
    "frommember": "BCEL",
    "fromuser": "2022456699",
    "fromuserfullname": "ADSAVIN THEPPHAKAN MR",
    "fromaccount": "010120000005477001",
    "tomember": "LVI",
    "toaccount": "1234567890",
    "totype": "ACCOUNT",
    "reference": "ABCDEFGHijklmn",
    "time": "2020-10-09 05:34:11"
  },
  "accountname": "ADSAVIN THEPPHAKAN",
  "accountccy": "LAK,USD"
}
```

5.3. Outgoing Transfer/Payment

5.3.1. Request to send to LAPNET

5.3.1.1. URL

- Development: <http://<LAPNET-IP>:8055/transfer>

- Production: **http://<LAPNET-IP>:8055/transfer**

5.3.1.2. Header

Content-Type: application/json

X-Source-Request-Signature: <<source Request signature>>

5.3.1.3. Body

Field	Type	Max-Length	Description
frommember	String	16	E.g. BCEL, LDB, APB, LVI
fromuser	String	64	E.g. narin@gmail.com
fromuserfullname	String(English)	256	E.g. ADSAVIN THEPPHAKAN MR
fromaccount	String	32	E.g. 010120000123456001
totype	String	16	ACCOUNT/CARD/QR
toaccount	String	2048	Account number or card number or full QR data
tomember	String	16	Only required if totype is ACCOUNT
reference	String	64	Member-generated unique reference. Should be the same as inquiry reference.
time	String	25	e.g: 2020-10-09 05:34:11
amount	Number (24,2)		If the transaction has a fixed amount, this amount must be the same as the amount received when inquired.
fee	Number (24,2)		Fee calculated by member according to "feelist". LAPNET will double-check this, if the fee not matched, LAPNET will reject with result FEE_NOT_CORRECT
ccy	String	3	Settlement CCY. If sender specifies USD, sending bank will need to settle in USD
purpose	String	512	If totype is ACCOUNT, the purpose is required. Purpose's requirement can be overridden by field purpose required when inquired.

5.3.1.4. Example

```
{
  "frommember": "BCEL",
  "fromuser": "2078729999",
  "fromuserfullname": "ADSAVIN THEPPHAKAN MR",
  "fromaccount": "010120000005477001",
  "totype": "ACCOUNT",
  "toaccount": "1234567890",
  "tomember": "LVI",
  "reference": "ABCDEFGHIJKLMN",
  "time": "2020-10-09 05:34:11",
  "amount": 100000,
  "ccy": "LAK",
  "purpose": "SOMETHING",
}
```

5.3.2. Response coming from LAPNET

5.3.2.1. Header

Content-Type: application/json

X-Source-Request-Signature: <<source request signature>>

X-LAPNET-Request-Signature: <<LAPNET request signature>>

X-Destination-Response-Signature: <<destination response signature>>

X-LAPNET-Response-Signature: <<LAPNET response signature>>

X-LAPNET-Transaction-ID: <<Sequence transaction ID generate by LAPNET>>

5.3.2.2. Body

Field	Type	Max-Length	Description
result	String	64	See Result Table
reference	String	64	Original request reference
shouldrevert	Integer 0 or 1		When the result is not "OK", if "shouldrevert" is 1 then refund the customer. Otherwise, follow the procedure when transfer fails. Use Integer instead of Boolean to make database record easier
time	String	25	e.g: 2020-10-09 05:34:11
originalmessage	JSON object		Original request message

5.3.2.3. Example

```
{
  "result": "OK",
  "reference": "ABCDEFGHijklmn",
  "time": "2020-10-09 05:34:11",
  "originalmessage": {
    "messagetype": "REQUEST",
    "frommember": "BCEL",
    "fromuser": "2078729999",
    "fromuserfullname": "ADSAVIN THEPPHAKAN MR",
    "fromaccount": "010120000005477001",
    "totype": "ACCOUNT",
    "toaccount": "1234567890",
    "tomember": "LVI",
    "reference": "ABCDEFGHijklmn",
    "time": "2020-10-09 05:34:11",
    "amount": 100000,
    "ccy": "LAK",
    "purpose": "SOMETHING",
  }
}
```

5.4. Incoming Transfer/Payment

5.4.1. Request received from LAPNET

5.4.1.1. URL

Members will need to provide the URL to LAPNET

5.4.1.2. Header

Content-Type: application/json

X-Source-Request-Signature: <<source Request signature>>

X-LAPNET-Request-Signature: <<LAPNET Request signature>>

X-LAPNET-Transaction-ID: <<Sequence transaction ID generate by LAPNET>>

5.4.1.3. Body

Field	Type	Max-Length	Description
-------	------	------------	-------------

frommember	String	16	E.g. BCEL, LDB, APB, LVI
fromuser	String	64	E.g. narin@gmail.com
fromuserfullname	String(English)	256	ADSAVIN THEPPHAKAN MR
fromaccount	String	32	E.g. 010120000123456001
totype	String	16	ACCOUNT/CARD/QR
toaccount	String	2048	Account number or card number or full QR data
tomember	String	16	Only required if totype is ACCOUNT
reference	String	64	Member-generated unique reference. Should be the same as inquiry reference.
time	String	25	e.g: 2020-10-09 05:34:11
amount	Number (24,2)		If the transaction has a fixed amount, this amount must be the same as the amount received when inquired.
fee	Number (24,2)		Fee calculated by member according to "feelist". LAPNET already double-check this
ccy	String	3	Settlement CCY. If sender specifies USD, sending bank will need to settle in USD
purpose	String	512	If totype is ACCOUNT, the purpose is required. Purpose's requirement can be overridden by field purposerequired when inquired.

5.4.1.4. Example

```
{
  "frommember": "BCEL",
  "fromuser": "2078729999",
  "fromuserfullname": "ADSAVIN THEPPHAKAN MR",
  "fromaccount": "010120000005477001",
  "totype": "ACCOUNT",
  "toaccount": "1234567890",
  "tomember": "LVI",
  "reference": "ABCDEFGHijklmn",
  "time": "2020-10-09 05:34:11",
  "amount": 100000,
  "ccy": "LAK",
  "purpose": "SOMETHING",
}
```

5.4.2. Response to send to LAPNET

5.4.2.1. Header

Content-Type: application/json

X-Source-Request-Signature: <<source request signature>>

X-LAPNET-Request-Signature: <<LAPNET request signature>>

X-Destination-Response-Signature: <<destination response signature>>

X-LAPNET-Transaction-ID: <<Sequence transaction ID generate by LAPNET>>

5.4.2.2. Body

Field	Type	Max-Length	Description
result	String	64	See Result Table
shouldrevert	Integer 1 or 0		Tell the source member whether to refund the customer or not when the result is not "OK". Use Integer instead of Boolean to make database record easier

reference	String	64	Original request reference
time	String	25	E.g. 2020-10-09 05:34:11
originalmessage	JSON object		Original request message

5.4.2.3. Example

```
{
  "result": "OK",
  "reference": "ABCDEFGHJKLMN",
  "time": "2020-10-09 05:34:11",
  "originalmessage": {
    "messagetype": "REQUEST",
    "frommember": "BCEL",
    "fromuser": "2078729999",
    "fromuserfullname": "ADSAVIN THEPPHAKAN MR",
    "fromaccount": "010120000005477001",
    "totype": "ACCOUNT",
    "toaccount": "1234567890",
    "tomember": "LVI",
    "reference": "ABCDEFGHJKLMN",
    "time": "2020-10-09 05:34:11",
    "amount": 100000,
    "ccy": "LAK",
    "purpose": "SOMETHING",
  }
}
```

5.5. Incoming Transaction Notification

5.5.1. Request coming from LAPNET

If a member's response is not correct or completed, LAPNET will notify via this channel. For example:

- X-Destination-Response-Signature is not correct
- Cannot contact source member

In the future, this channel will be used to notify transaction status for Merchant-Scan-Customer QR payment

5.5.1.1. URL

Member will need to provide the URL

5.5.1.2. Header

Content-Type: application/json

X-LAPNET-Request-Signature: <<LAPNET request signature>>

X-LAPNET-Transaction-ID: <<Sequence transaction ID generate by LAPNET>>

5.5.1.3. Body

Field	Type	Max-Length	Description
result	String	64	See Result Table
reference	String	64	Original message's reference
transactionid	Number		Original LAPNET's transaction ID
time	String	25	ISO 8601. E.g. 2020-10-09T05:34:11+07:00
originalmessage	JSON object		Original response message that has a problem

5.5.1.4. Example

```
{
  "result": "OK",
  "reference": "YI3NZK83V08V",
  "time": "2022-07-14 21:13:55",
  "originalmessage": {
    "transactionid": 14898519,
    "result": "CANNOT_CONNECT_TO_DESTINATION",
    "time": "2022-07-14 21:13:54",
    "reference": "YI3NZK83V08V"
  }
}
```

6. Check Transaction Status

Members can get transaction details recorded on LAPNET to compare with their own records. For example, when connection is lost, and members are not sure whether the transaction is completed or not.

Inquiry and transfer requests will be recorded when LAPNET successfully parsed the requests.

If "transferresponse.result" is empty, that means transaction is not completed yet, it could fail or success, wait for a while and check transaction again before decide to refund or not.

6.1.1. Request to send to LAPNET

6.1.1.1. URL

- Development: **http://<LAPNET-IP>:8055/report/transaction**
- Production: **http://<LAPNET-IP>:8055/report/transaction**

6.1.1.2. Header

Content-Type: application/json

X-Source-Request-Signature: <<source Request signature>>

6.1.1.3. Body

Field	Type	Max-Length	Description
frommember	String	16	E.g. BCEL, LDB, APB, LVI
reference	String	64	E.g. YNYJK5QZUB8FW1C1IZSE

6.1.1.4. Example

```
{
  "frommember": "BCEL",
  "reference": " YNYJK5QZUB8FW1C1IZSE",
}
```

6.1.2. Response coming from LAPNET

6.1.2.1. Header

Content-Type: application/json

X-Source-Request-Signature: <<source request signature>>

X-LAPNET-Response-Signature: <<LAPNET response signature>>

6.1.2.2. Body Example

```
{
  "inquiryrequest": {
```

```

    "id": 21116,
    "frommember": "LVB",
    "fromuser": "LVB",
    "fromaccount": "",
    "totype": "ACCOUNT",
    "toaccount": "010120000005477001",
    "tomember": "BCEL",
    "reference": "YNYJK5QZUB8FW1C1IZSE",
    "time": "2020-11-20 14:37:10"
  },
  "inquiryresponse": {
    "id": 21116,
    "accountname": "ADSAVIN THEPPHAKAN MR",
    "accountccy": "LAK",
    "result": "OK",
    "time": "2020-11-20 14:36:58"
  },
  "transferrequest": {
    "id": 21122,
    "frommember": "LVB",
    "fromuser": "NGUYEN TRIEU NAM MR",
    "fromaccount": "01000110043713",
    "totype": "ACCOUNT",
    "toaccount": "010120000005477001",
    "tomember": "BCEL",
    "reference": "YNYJK5QZUB8FW1C1IZSE",
    "time": "2020-11-20 14:59:35",
    "amount": 10000.0,
    "fee": 1000,
    "ccy": "LAK",
    "purpose": "Transfer from LVB account: 01000110043713 to BCEL account:
010120000005477001"
  },
  "transferresponse": {
    "result": "ACCOUNT_CLOSED",
    "shouldrevert": true,
    "time": "2020-11-20 14:59:35"
  }
}

```

7. Settlement Report

Get daily summary and transactions of successful transfer transactions that members need to settle.

Member can only get settlement report of passed days, cannot view current's day

Transactions range is based on request time, including and after 00:00 of specified day, but not before 00:00 of next day (requesttime >= 00:00 AND requesttime < 00:00 + 1 day)

Settlement items separated by CCY.

7.1.1. Request to send to LAPNET

7.1.1.1. URL

- Development: <http://<LAPNET-IP>:8055/report/settlement>

- Production: <http://<LAPNET-IP>:8055/report/settlement>

7.1.1.2. Header

Content-Type: application/json

X-Source-Request-Signature: <<source Request signature>>

7.1.1.3. Body

Field	Type	Max-Length	Description
frommember	String	16	E.g. BCEL, LDB, APB, LVI
date	String	10	E.g. 2020-11-21

7.1.1.4. Example

```
{
  "frommember": "BCEL",
  "date": "2020-11-21",
}
```

7.1.2. Response coming from LAPNET

7.1.2.1. Header

Content-Type: application/json

X-Source-Request-Signature: <<source request signature>>

X-LAPNET-Response-Signature: <<LAPNET response signature>>

7.1.2.2. Body Example

```
{
  "date": "2020-11-19",
  "summaries": [
    {
      "ccy": "LAK",
      "debit": 0.0,
      "credit": 15000.0,
      "balance": 15000.0,
      "issettled": 0
    },
    {
      "ccy": "USD",
      "debit": 0.0,
      "credit": 10.0,
      "balance": 10.0,
      "issettled": 0
    }
  ],
  "transactions": [
    {
      "id": 19029,
      "frommember": "JDB",
      "fromuser": "001074819",
      "fromaccount": "00120010003159383",

```

```

    "totype": "ACCOUNT",
    "toaccount": "010120100005477001",
    "tomember": "BCEL",
    "reference": "2020111911141911",
    "time": "2020-11-19 14:19:11",
    "amount": 5.0,
    "ccy": "USD",
    "purpose": "memo text"
  },
  {
    "id": 19031,
    "frommember": "JDB",
    "fromuser": "001063985",
    "fromaccount": "00120010002982794",
    "totype": "ACCOUNT",
    "toaccount": "010120000005477001",
    "tomember": "BCEL",
    "reference": "20201119142300",
    "time": "2020-11-19 14:23:00",
    "amount": 15000.0,
    "ccy": "LAK",
    "purpose": "memo text"
  },
  {
    "id": 21055,
    "frommember": "JDB",
    "fromuser": "001074819",
    "fromaccount": "00120010003159383",
    "totype": "ACCOUNT",
    "toaccount": "010120100005477001",
    "tomember": "BCEL",
    "reference": "2020112011141911",
    "time": "2020-11-19 14:19:11",
    "amount": 5.0,
    "ccy": "USD",
    "purpose": "memo text"
  }
],
"finlink": {
  "ROADTAX": [],
  "LANDTAX": [],
  "LNSW": [],
  "SMARTVAT": [],

```

```

"EASYTAX": [],
"FEE002": [],
"summaries": {
  "LANDTAX": 100,
  "ROADTAX": 100,
  "LNSW": 100,
  "SMARTVAT": 100,
  "EASYTAX": 100,
  "FEE002": 100
}
},
"crossborders": {
  "Cambodia":{
    "summaries": [
      {
        "member": "BCEL",
        "ccy": "KHR",
        "txdate": "2023-11-19 14:19:11",
        "inbound": "1",
        "inboundfee": "500",
        "outbound": "1",
        "outboundfee": "500",
        "debit": "1",
        "credit": "1",
        "balance": "20000",
        "issettled": "1",
        "settleid": "123"
      }
    ],
    "transactions": [
      {
        "id": 77382672,
        "txtime": "2023-12-08 09:23:17",
        "requesttime": "2023-12-08 09:23:17",
        "reference": "QR2308070803",
        "frommember": "CAMBODIA",
        "fromuser": "advin@bcel.com.la",
        "fromaccount": "010120000005477001",
        "totype": "QR",
        "toaccount":
"00020101021138570016A00526628466257701082771041802030020314mc

```

```

h18855A2CD035204737253034185802LA6002VT621202082222407163042A9
4",
  "tomember": "BCEL",
  "responsetime": "2023-12-08 09:23:17",
  "result": "OK",
  "amount": 40000,
  "ccy": "LAK",
  "purpose": "Testing",
  "isreconciled": 0,
  "issettled": 0,
  "settletime": null,
  "txtype": null,
  "fee": 0,
  "originalreference": null,
  "shouldrevert": 1,
  "qrtype": "002",
  "requesttime_dt": "2023-12-08"
},
{
  "id": 77382902,
  "txtime": "2023-12-08 09:24:04",
  "requesttime": "2023-12-08 09:24:04",
  "reference": "QR2308070804",
  "frommember": "BCEL",
  "fromuser": "t.rongroeung@gmail.com",
  "fromaccount": "010120000005477001",
  "totype": "QR",
  "toaccount":
"00020101021138570016A00526628466257701082771041802030020314mc
h18855A2CD035204737253034185802LA6002VT621202082222407163042A9
4",
  "tomember": "CAMBODIA",
  "responsetime": "2023-12-08 09:24:04",
  "result": "OK",
  "amount": 20000,
  "ccy": "LAK",
  "purpose": "Testing",
  "isreconciled": 0,
  "issettled": 0,
  "settletime": null,
  "txtype": null,
  "fee": 0,
  "originalreference": null,

```

```

        "shouldrevert": 0,
        "qrtype": "002",
        "requesttime_dt": "2023-12-08"
    }
]
},
"Thai_ITMX":{
    "summaries": [
        {
            "member": "BCEL",
            "ccy": "THB",
            "txdate": "2023-11-19 14:19:11",
            "inbound": "1",
            "inboundfee": "500",
            "outbound": "1",
            "outboundfee": "500",
            "debit": "1",
            "credit": "1",
            "balance": "20000",
            "issettled": "1",
            "settleid": "123"
        }
    ],
    "transactions": [
        {
            "id": 77382672,
            "txtime": "2023-12-08 09:23:17",
            "requesttime": "2023-12-08 09:23:17",
            "reference": "QR2308070803",
            "frommember": "Thai_ITMX",
            "fromuser": "advin@bcel.com.la",
            "fromaccount": "010120000005477001",
            "totype": "QR",
            "toaccount":
"00020101021138570016A0052662846625770108277104180203002
0314mch18855A2CD035204737253034185802LA6002VT62120208222
2407163042A94",
            "tomember": "BCEL",
            "responsetime": "2023-12-08 09:23:17",
            "result": "OK",
            "amount": 40000,
            "ccy": "LAK",
            "purpose": "Testing",
            "isreconciled": 0,

```

```

        "issettled": 0,
        "settletime": null,
        "txtype": null,
        "fee": 0,
        "originalreference": null,
        "shouldrevert": 1,
        "qrtype": "002",
        "requesttime_dt": "2023-12-08"
    },
    {
        "id": 77382902,
        "txtime": "2023-12-08 09:24:04",
        "requesttime": "2023-12-08 09:24:04",
        "reference": "QR2308070804",
        "frommember": "BCEL",
        "fromuser": "t.rongroeung@gmail.com",
        "fromaccount": "010120000005477001",
        "totype": "QR",
        "toaccount":
        "00020101021138570016A00526628466257701082771041802030020314mc
        h18855A2CD035204737253034185802LA6002VT621202082222407163042A9
        4",
        "tomember": "Thai_ITMX",
        "responsetime": "2023-12-08 09:24:04",
        "result": "OK",
        "amount": 20000,
        "ccy": "LAK",
        "purpose": "Testing",
        "isreconciled": 0,
        "issettled": 0,
        "settletime": null,
        "txtype": null,
        "fee": 0,
        "originalreference": null,
        "shouldrevert": 0,
        "qrtype": "002",
        "requesttime_dt": "2023-12-08"
    }
]
},
,
"CHINA_UPI":{
    "summaries": [

```

```

{
  "member": "BCEL",
  "ccy": "CNY",
  "txdate": "2023-11-19 14:19:11",
  "inbound": "1",
  "inboundfee": "500",
  "outbound": "1",
  "outboundfee": "500",
  "debit": "1",
  "credit": "1",
  "balance": "20000",
  "issettled": "1",
  "settleid": "123"
}],
"transactions": [
  {
    "id": 77382672,
    "txtime": "2023-12-08 09:23:17",
    "requesttime": "2023-12-08 09:23:17",
    "reference": "QR2308070803",
    "frommember": "CHINA_UPI",
    "fromuser": "advin@bcel.com.la",
    "fromaccount": "010120000005477001",
    "totype": "QR",
    "toaccount":
      "00020101021138570016A0052662846625770108277104180203002
      0314mch18855A2CD035204737253034185802LA6002VT62120208222
      2407163042A94",
    "tomember": "BCEL",
    "responsetime": "2023-12-08 09:23:17",
    "result": "OK",
    "amount": 40000,
    "ccy": "LAK",
    "purpose": "Testing",
    "isreconciled": 0,
    "issettled": 0,
    "settletime": null,
    "txtype": null,
    "fee": 0,
    "originalreference": null,
    "shouldrevert": 1,
    "qrtype": "002",
    "requesttime_dt": "2023-12-08"
  }
]

```

```

    },
    {
      "id": 77382902,
      "txtime": "2023-12-08 09:24:04",
      "requesttime": "2023-12-08 09:24:04",
      "reference": "QR2308070804",
      "frommember": "BCEL",
      "fromuser": "t.rongroeung@gmail.com",
      "fromaccount": "010120000005477001",
      "totype": "QR",
      "toaccount":
        "00020101021138570016A00526628466257701082771041802030020314mc
        h18855A2CD035204737253034185802LA6002VT621202082222407163042A9
        4",
      "tomember": "CHINA_UPI",
      "responsetime": "2023-12-08 09:24:04",
      "result": "OK",
      "amount": 20000,
      "ccy": "LAK",
      "purpose": "Testing",
      "isreconciled": 0,
      "issettled": 0,
      "settletime": null,
      "txtype": null,
      "fee": 0,
      "originalreference": null,
      "shouldrevert": 0,
      "qrtype": "002",
      "requesttime_dt": "2023-12-08"
    }
  ]
}
}
}

```

8. Currency Reimbursement Report

The Currency Reimbursement Report is a simple tool that helps a service bank understand how much foreign currency they will receive from a settlement bank based on the actual transactions made.

The main purpose of this report is to provide an easy-to-understand figure that shows the total amount of foreign currency the service bank will be reimbursed by the settlement bank, based on the actual transaction amounts. This helps the bank keep track of its foreign currency earnings and manage its finances more effectively..

8.1.1. Request to send to LAPNET

8.1.1.1. URL

- Development: **http://<LAPNET-IP>:8055/report/reimbursment**
- Production: **http://<LAPNET-IP>:8055/report/reimbursment**

8.1.1.2. Header

Content-Type: application/json

X-Source-Request-Signature: <<source Request signature>>

8.1.1.3. Body

Field	Type	Max-Length	Description
frommember	String	16	E.g. BCEL, LDB, APB, LVI
month	String	10	E.g. 2020-11

8.1.1.4. Example

```
{
  "frommember": "BCEL",
  "date": "2020-11",
}
```

8.1.2. Response coming from LAPNET

8.1.2.1. Header

Content-Type: application/json

X-Source-Request-Signature: <<source request signature>>

X-LAPNET-Response-Signature: <<LAPNET response signature>>

8.1.2.2. Example response

```
{
  "member": "BCEL"
  "reimbursement_sum": {
    "Cambodia":{"amount": 5000, "ccy": "KHR"},
    "Thai_ITMX":{"amount": 5000, "ccy": "THB"},
  }
}
```

9. How to proceed when transfer failed

9.1. For source member

- When the **"result"** is **"OK"** the transfer is completed.
- When the **"result"** is **not "OK"** (including empty/null) and **"shouldrevert"** if it is **true**, source member must refund the customer.
- When **"result"** is **not "OK"** (including empty/null) and **"shouldrevert"** is **false or missing**, source member **MUST NOT REFUND** the customer. Source member must periodically check for transaction result using function "Check Transaction Status" to get **transferresponse.shouldrevert** if it is **true** then refund the customer. Otherwise, continue to periodically check until before the settlement time. After the settlement, the transaction should be resolved manually.

9.2. For destination member

- If the transfer is successful, **"result"** must be **"OK"**.

- If the transfer failed and destination account is not credited, **"result"** must be the reason of a failure (E.g., ACCOUNT_BLOCKED, CCY_INCORRECT, LIMIT_EXCEEDED) and **"shouldrevert"** must be **true**.
- If the transfer failed and destination account might be credited (E.g., core banking timeout, network issue) **"result"** must be the reason of a failure and **"shouldrevert"** must be **false**. Member must periodically check its own destination account to confirm if it has been credited or not. If credited, mark the **"result"** as **"OK"**, if not credited, mark **"shouldrevert"** as **true**.

9.3. For mobile banking app

- When performing a transfer, after receiving **"result"** as **not "OK"**, check for **"shouldrevert"**, if it is **true**, tell the customer that the transaction is not successful, the reason is according to **"result"** code.
- If **"result"** is **not "OK"** and **"shouldrevert"** is **false** or missing, tell the customer that "transaction is taking longer than usual, please check back later" or "we will notify you after the transaction is completed" while in the backend, periodically check for transaction status to see if **"result"** is **"OK"** or **"shouldrevert"** is **true** then proceed and notify customer accordingly.

"result" format should be uppercase and use underscore instead of space. "result" should not be unique for each account. E.g., it should be ACCOUNT_BLOCKED instead of ACCOUNT_010123456_BLOCKED.

10. Transaction Signature

1. Generate signature using SHA256withRSA algorithm, using HTTP body as data and private key as a key. E.g.:

```
KeyFactory keyFactory = KeyFactory.getInstance("RSA");
Signature sig = Signature.getInstance("SHA256withRSA");
sig.initSign(keyFactory.generatePrivate(new PKCS8EncodedKeySpec(privateKey)));
sig.update(data);
return sig.sign();
```
2. Convert signature into HEX format and put in header according to message spec.

11. Reference Generation

Reference field must be 20 digits of uppercase alphanumeric (allowed characters are A, B, C, D, E, F, G, H, I, J, K, L, M, O, P, Q, R, S, T, U, V, W, X, Y, Z, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9)

For example: KJ345H3JK4H53KL45H28

Chance of duplication is 1 in 13,367,494,538,843,734,067,838,845,976,576. In the event duplication, member must generate a new reference

Sample Java code:

```
// Generator can be static
static RandomStringGenerator referenceGenerator = new RandomStringGenerator.Builder()
    .selectFrom("ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789").toCharArray())
    .usingRandom(new SecureRandom()::nextInt)
    .build();
//Use this to generate reference anywhere
referenceGenerator.generate(20)
```

12. Connectivity

Allow Firewall to white listed Production's Server IPs 175.28.1.3, 175.28.1.4, 175.28.1.5 and UAT's server IP 175.28.1.5 both Environment running under Port 8055

13. LMPS Side Testing Checklist

Pass	Failed	Description	Remark
------	--------	-------------	--------

☐	☐	Send Request to http://<MEMBER-IP>:8055/inquiry	
☐	☐	Send Request to http://<MEMBER-IP>:8055/transfer	
☐	☐	Send Request to http://<MEMBER-IP>:8055/notify	

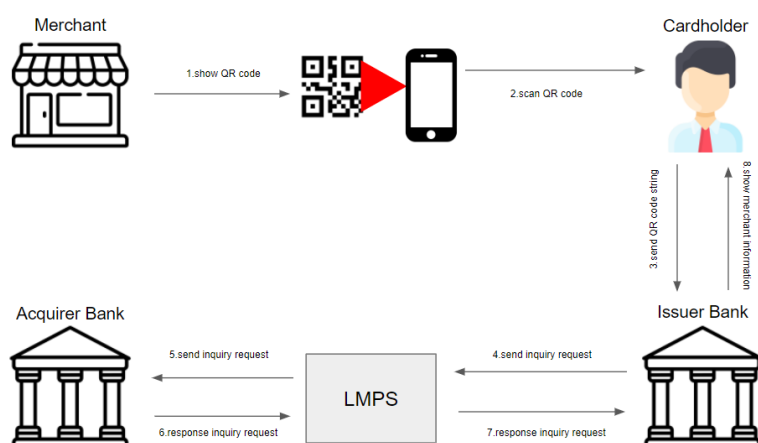
14. Member Side Testing Checklist

Pass	Failed	Description	Remark
☐	☐	Send Request to http://<LAPNET-IP>:8055/inquiry	
☐	☐	Send Request to http://< LAPNET-IP >:8055/transfer	
☐	☐	Send Request to http://< LAPNET-IP >:8055/notify	

15. Lao National QR code

15.1.Lao QR code Inquiry Project

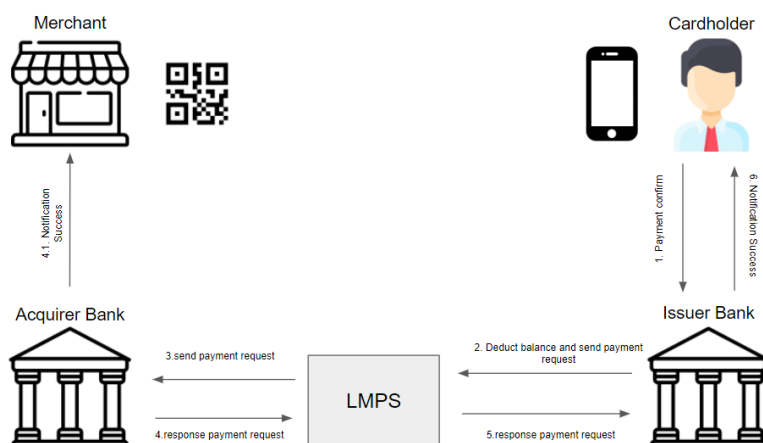
Inquiry Process



1. Merchant show Static or dynamic QR code
2. Consumer scan QR code by mobile application then click “Scan”
3. Mobile application send request to Issuer bank
4. Issuer bank Check QR code Type Laos, UPI, THAI, Vietnam send inquiry over LMPS to Acquire bank only no data verification
5. Acquire bank get inquiry
6. Acquire bank check QR and merchant’s information
7. Acquire response to Issuer bank over LMPS
8. Issuer bank response Merchant name, price, Fee show on Mobile application

15.2.Lao QR payment confirmation process

Payment Process



1. After consumer check merchant information click “Confirm”, Mobile application send request to Issuer bank
2. Issuer bank debit balance and send payment success to Acquire bank over LMPS
3. Acquire bank get confirm request and credit balance to merchant account (If requested by UnionPay “CHINA_UPI”, it must only be completed)
4. Acquire bank send notification to Merchant and response process to LMPS
5. LMPS response to Issuer bank
6. Issuer bank send payment information and notification to consumer

15.3.QR code structure

Example for Static QR

ID	Data Object	Type	Length	Presence	Description or Example
00	Payload format Indicator	N	02	M	01
01	Point of Initiation method	N	02	O	12
15	UnionPay Tag	Ans	Var up to 35	M	Acquirer IIN + Forwarding IIN + Merchant ID
	Acquirer IIN	Ans	Var up to 8	M	20310418
	Forwarding IIN	Ans	Var up to 8	M	00010344
	Merchant ID	Ans	Var up to 15	M	ABCDE123456789 (same as Receiver ID but limit 15 charactor)
38	Lao National QRC for QR payment	Ans			
00	AID	Ans	16	M	A005266284662577
01	<u>IIN</u>	N	08	M	27710418 detail
02	<u>Payment type</u>	N	03	M	002 detail
03	Receiver ID	Ans	Var up to 25	M	ABCDE123456789
52	<u>Merchant Category code</u>	N	04	M	5813
53	Transaction Currency	Ans	03	M	418 (LAK)
58	Country Code	Ans	02	M	LA

59	Merchant Name	Ans	Var up to 25	M	Must provide every time
60	Merchant city	Ans	Var up to 15	M	Vientiane
63	Cyclic Redundancy Check		04	M	XXXX CRC implement

Example for Dynamic QR

ID	Data Object	Type	Length	Presence	Description or Example
00	Payload format Indicator	N	02	M	01
01	Point of Initiation method	N	02	O	12
15	UnionPay Tag	Ans	Var up to 35	M	Acquirer IIN + Forwarding IIN + Merchant ID
	Acquirer IIN	Ans	Var up to 8	M	20310418
	Forwarding IIN	Ans	Var up to 8	M	00010344
	Merchant ID	Ans	Var up to 15	M	ABCDE123456789 (same as Receiver ID but limit 15 charactor)
38	Lao National QRC for QR payment	Ans			
00	AID	Ans	16	M	A005266284662577
01	IIN	N	08	M	27710418 detail
02	Payment type	N	03	M	002 detail
03	Receiver ID	Ans	Var up to 25	M	ABCDE123456789
52	Merchant Category code	N	04	M	5813
53	Transaction Currency	Ans	03	M	418 (LAK)
54	Transaction Amount	N	Var up to 13	C	250000
58	Country Code	Ans	02	M	LA
59	Merchant Name	Ans	Var up to 25	M	Must provide every time
60	Merchant city	Ans	Var up to 15	M	Vientiane
62	Additional Data Field Template	Ans	Var up to 99	O	
01	Bill Number	Ans	Var up to 25	O	00000100
02	Mobile number	Ans	Var up to 25	O	7XXXXXXX
03	Store Label	Ans	Var up to 25	O	SYHOME
04	Loyalty number	Ans	Var up to 25	O	
05	Reference Label	Ans	Var up to 25	O	00000021
06	Customer Label	Ans	Var up to 25	O	
07	Terminal Label	Ans	Var up to 25	O	T00001
08	Purpose of Transaction	Ans	Var up to 25	O	Iced coffee
09	Additional Consumer Data Request	Ans	Var up to 3	O	
63	Cyclic Redundancy Check		04	M	XXXX CRC implement

***Remark:**

About ID 52,59,60 Use for Merchant QR Only

- N Numeric

- Ans Alphabetic, numeric and special characters.

- M Mandatory

- O Optional

- C Conditional

15.4.Member IIN number

No.	Bank Name	38.01 IIN
1	ACLE	70080418
2	APB	70020418
3	BCEL	27710418
4	BIC	70110418
5	BOC	70100418
6	ICBC	70070418
7	IDB	70140418
8	JDB	32170418
9	LDB	70030418
10	LVB	70050418
11	MJB	70040418
13	SACOM	70130418
14	STB	70150418
15	KBank	70170418
16	VTB	70120418
17	BFL	37160418

15.5.Payment type

38.02 Payment Type	Name
001	Credit fund transfer
002	QR Payment
003	Bill payment
004	E-donation
005	E-Commerce
006	Electric Payment
007	Water Payment
009	Government pay
006-999	Reserved for Future Use

15.6. Merchant Category type

MCC	Description
-----	-------------

4900	Electric, Gas, Sanitary and Water Utilities
9311	Tax Payments

16. CRC implement

The checksum shall be calculated according to [ISO/IEC 13239] using the polynomial '1021' (hex) and initial value 'FFFF' (hex). The data over which the checksum is calculated shall cover all data objects, including their ID, Length and Value, to be included in the QR Code, in their respective order, as well as the ID and Length of the CRC itself (but excluding its Value).

Algorithm: CRC-16/CCITT-FALSE

input: ASCII

output: HEX

Initial: 0Xffff

QR_STRUCTURE = [(TAG01+ LENGTH+DATA) + (TAG01+LENGTH +DATA)+...(TAG62+LENGTH+DATA)]

CRC = CRC_16_CCITT_FALSE_ENCRYPT (QR_STRUCTURE + 6304)

EMVCO_QR_STRING = (QR_CODE_STRUCTURE + 6304 + CRC)

Example:

Generate string QR code with Lao QR structure and add last CRC tag 63 by add 6304 and leave CRC value, then use String QR code below to

generate: "00020101021138590016A00526628466257701082771041802030020316mch5949fa044ed9d53034185802LA5910Bunny test6003VTE6304". Then will get CRC is 62F5 and put it on the last

17. Thailand Cross border payment

17.1. Exchange THB API for Service Bank

Service Bank can setting rate for THB only. Do not specify other currency other than specified

17.1.1 Exchange Rates Thailand

17.2.1.1 Get Exchange Rates

- URL: <http://<LMPS-IP>:8055/crossborder/getexchangerates>

17.1.1.1.1 Request Body

```
{
  "frommember": "",
  "time": "2023-04-27 14:08:35",
  "reference": "HEZFZH901NDWYTWDOMY5",
  "ccy": "THB"
```

```
}
```

17.1.1.1.2 Response Body

```
{
  "reference": "IE3FAG73JLDLWWLDU4",
  "time": "2023-09-26 15:44:03",
  "result": "OK",
  "originalmessage": {
    "frommember": "",
    "reference": "IE3FAG73JLDLWWLDU4",
    "ccy": "THB",
    "time": "2023-09-21 08:54:36"
  },
  "exchangerates": [
    {
      "ccy": "THB",
      "lastupdate": "2023-09-26 20:35:39",
      "rate": "520.8"
    },
    ...
  ]
}
```

17.1.1.2 Register/Update Exchange Rate

- URL: <http://<LMPS-IP>:8055/crossborder/exchangerate>

17.1.1.2.1 Request Body

```
{
  "frommember": "",
  "reference": "IE3FAG73JLDLWWLDU4",
  "ccy": "THB",
  "rate": "520.8",
  "time": "2023-09-21 08:54:36"
}
```

17.1.1.2.2 Response Body

```
{
  "reference": "IE3FAG73JLDLWWLDU4",
  "time": "2023-09-26 15:46:38",
  "result": "OK",
  "originalmessage": {
    "frommember": "",
    "reference": "IE3FAG73JLDLWWLDU4",
    "ccy": "THB",
    "rate": "520.8",
    "time": "2023-09-21 08:54:36"
  }
}
```


17.2. Outbound Incoming Transfer (Thai-Lao)

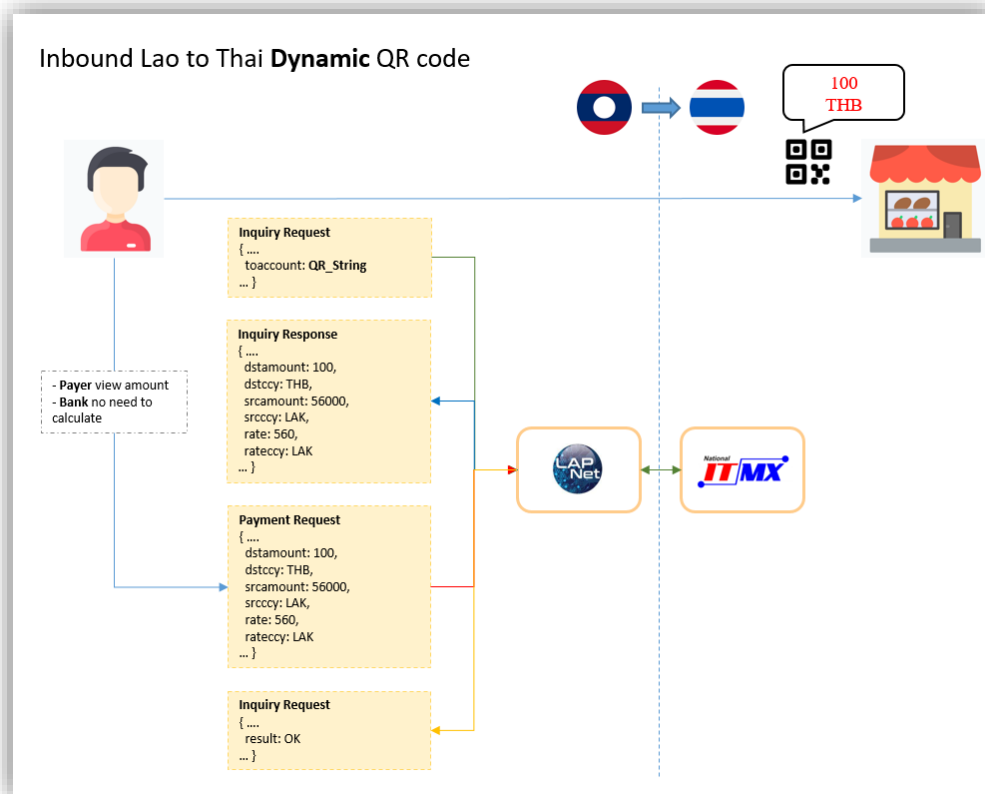
When requested by Thailand is must be Follow up process Incoming Transfer domestic

17.3. Inbound Lookup request (Lao-Thai)

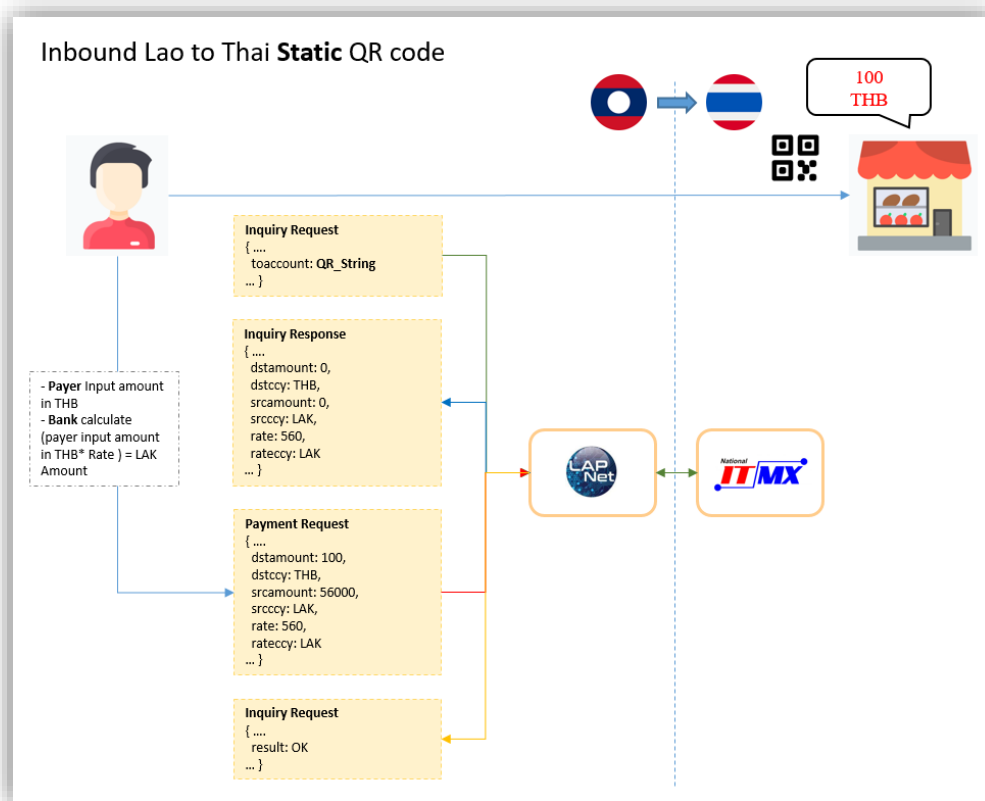
Please check if the QR code contains the string 'A000000677010112'. If it does, or if tag 30 is found and tag 30.00 contains 'A000000677010112', please forward that QR code to the LMPS service. Additionally, please add the user's full name to field 'fromuserfullname' the message.

17.4. Lookup and payment flow

In case Lao scan Thai **Dynamic** QR code system get amount THB in QR code already LAPNet will return "dstamount", "srcamount" and "rate" on response of lookup. On payer screen just show amount in Destination amount and amount in LAK.



In case Lao scan Thai **Static** QR code system don't get amount. LAPNet will return "dstamount=0", "srcamount=0" and "rate" on response of lookup. On payer screen need to input amount in THB and Member need to calculate "srcamount" = ("dstamount" * "rate") and show both amount source amount and destination amount on payer confirm screen.



17.4.1. URL

- Development: <http://<LAPNET-IP>:8055/inquiry>
- Production: <http://<LAPNET-IP>:8055/inquiry>

17.4.2. Body

Field	Type	Max-Length	Description
frommember	String	16	E.g. BCEL, LDB, APB, LVI, CHINA_UPI
fromuser	String	64	E.g. narin@gmail.com
fromuserfullname	String(English)	256	ADSAVIN THEPPHAKAN MR
fromaccount	String	32	E.g. 010120000123456001
totype	String	16	ACCOUNT/CARD/QR
toaccount	String	2048	Account number or card number or full QR data
reference	String	64	Member-generated unique reference. Should be the same as inquiry reference.
time	String	25	e.g. 2020-10-09 05:34:11

17.4.3. Example

Inquiry request message:

```
{
  "frommember": "APB",
  "fromuser": "lex@gmail.com",
```

```

    "fromuserfullname": "Lex laolapnet",
    "fromaccount": "012000387021000021",
    "toaccount":
"00020101021230810016A00000067701011201150105536092641990215000002205500093031945009901121021
000015303764540520.005802TH5911BBL TEST 1062120708450099016304B062",
    "totype": "QR",
    "reference": "112233445566",
    "time": "2023-07-30 10:08:28"
}

```

Inquiry response:

```

{
  "result": "OK",
  "reference": "112233445566",
  "time": "2023-07-05 09:55:04.191",
  "originalmessage": {
    "frommember": "APB",
    "fromuser": "lex@gmail.com",
    "fromuserfullname": "Lex laolapnet",
    "fromaccount": "012000387021000021",
    "toaccount":
"00020101021230810016A00000067701011201150105536092641990215000002205500093031945009901121021
000015303764540520.005802TH5911BBL TEST 1062120708450099016304B062",
    "totype": "QR",
    "reference": "112233445566",
    "time": "2023-07-30 10:08:28"
  },
  "accountname": "TESCO LOTUS CO.,LTD.",
  "amount": "0",
  "dstamount": "100",
  "dstccy": "THB",
  "srcamount": "56000",
  "srcccy": "LAK",
  "rate": "560",
  "rateccy": "LAK"
}

```

17.5. Inbound payment request (Lao-Thai)

After receiving a response from LMPS, the accountccy and rate will be provided. If accountccy is THB, it means that the customer has input an amount in THB. The rate is used to convert the amount that the customer input into THB to LAK. For example, if accountccy=THB and rate=520, and the customer inputs 10 THB, the amount will be 10×520 , which equals 5200 LAK.

17.5.1. URL

- Development: `http://<LAPNET-IP>:8055/transfer`
- Production: `http://<LAPNET-IP>:8055/transfer`

17.5.2. Body

Field	Type	Max-Length	Description
frommember	String	16	E.g. BCEL, LDB, APB, LVI
fromuser	String	64	E.g. narin@gmail.com
fromuserfullname	String(English)	256	ADSAVIN THEPPHAKAN MR
fromaccount	String	32	E.g. 010120000123456001
totype	String	16	ACCOUNT/CARD/QR
toaccount	String	2048	Account number or card number or full QR data
reference	String	64	Member-generated unique reference. Should be the same as inquiry reference.
time	String	25	e.g.: 2020-10-09 05:34:11
amount	Number (24,2)		Transaction Amount THB x Rate
fee	Number (24,2)		Fee calculated by member according to "feelist". LAPNET already double-check this
ccy	String	3	Settlement CCY. If sender specifies USD, sending bank will need to settle in USD
purpose	String	512	If totype is ACCOUNT, the purpose is required. Purpose's requirement can be overridden by field purpose required when inquired.
dstamount	Number (24,2)		Destination amount: Amount for payment to merchant, if 0 is mean static QR code need to input amount on payer screen in dstccy
dstccy	String	3	Destination ccy: this example is Payment to Thai is dstccy = THB
srcamount	Number (24,2)		Source amount: Amount for deduct to Lao payer, 0 is need to get input from payer and calculate with Rate
srcccy	String	3	Source ccy: Lao is LAK
rate	Number (24,2)		Rate: Exchange from LAPNet in Source ccy buy Destination ccy, this is Lao-Thai is 560
rateccy	String	3	Rate ccy: LAK

17.5.3. Payment Example

Payment request

```
{
  "frommember": "APB",
  "fromuser": "lex@gmail.com",
  "fromuserfullname": "lex laolapnet",
```

```

    "fromaccount": "012000387021000021",
    "toaccount":
"00020101021230810016A00000067701011201150105536092641990215000002205500093031945009901121021
000015303764540520.005802TH5911BBL TEST 1062120708450099016304B062",
    "totype": "QR",
    "reference": "112233445566",
    "time": "2023-07-30 10:08:28"
    "amount": "56000",
    "fee": 0.0,
    "ccy": "LAK",
    "purpose": "NO SECOND NAME|1234|12345|11231859",
    "dstamount": "100",
    "dstccy": "THB",
    "srcamount": "56000",
    "srcccy": "LAK",
    "rate": "560",
    "rateccy": "LAK"
}

```

Payment response

```

{
    "result": "OK",
    "reference": "230705021141BCEL9909",
    "time": "2023-07-06 14:34:33.856",
    "originalmessage": {
        "frommember": "APB",
        "fromuser": "lex@gmail.com",
        "fromuserfullname": "Lex laolapnet",
        "fromaccount": "012000387021000021",
        "toaccount":
"00020101021230810016A00000067701011201150105536092641990215000002205500093031945009901121021
000015303764540520.005802TH5911BBL TEST 1062120708450099016304B062",
        "totype": "QR",
        "reference": "112233445566",
        "time": "2023-07-30 10:08:28"
        "amount": "56000",
        "fee": 0.0,
        "ccy": "LAK",
        "purpose": "NO SECOND NAME|1234|12345|11231859",
        "dstamount": "100",
        "dstccy": "THB",
        "srcamount": "56000",
    }
}

```

```

    "srccy": "LAK",
    "rate": "560",
    "rateccy": "LAK"
  }
}

```

18. China Cross border payment

18.1. UnionPay QR

This section describes the QR code structure for supporting receiving payments from UnionPay. It is necessary to add Tag 15, which is reserved for UnionPay.

18.1.1 Static QRC

- Tag ID 52, the Merchant Category Code can be used by the acquirer or UnionPay if they use it themselves. The acquirer needs to convert/match the code to the UnionPay code at the system backend for transaction processing. Please refer to the file titled "Updated UPI MCC List (Including Newly-added MCCs).pdf" for the MCC information.
- Tag ID 53, the Transaction Currency will only use the LAK code, which is 418.
- Tag ID 54, leave the Transaction Amount tag empty, as the mobile app will prompt the user to enter the amount for the QRC transaction.
- Tag 59 is the Merchant Name, which is mandatory for UnionPay. Please raise this as a requirement to add.

Example for Static QRC

ID	Data Object	Type	Length	Presence	Description or Example
00	Payload format Indicator	N	02	M	01
01	Point of Initiation method	N	02	O	12
15	UnionPay Tag	Ans	Var up to 35	M	Acquirer IIN + Forwarding IIN + Merchant ID
	Acquirer IIN	Ans	Var up to 8	M	00010344
	Forwarding IIN	Ans	Var up to 8	M	00010344
	Merchant ID	Ans	Var up to 15	M	ABCDEFGHIJKLMNO
38	Lao National QRC for QR payment	Ans			
00	AID	Ans	16	M	A005266284662577
01	<u>IIN</u>	N	08	M	27710418 detail
02	<u>Payment type</u>	N	03	M	002 detail
03	Receiver ID	Ans	Var up to 25	M	ABCDE123456789
52	<u>Merchant Category code</u>	N	04	M	5813
53	Transaction Currency	Ans	03	M	418 (LAK)
58	Country Code	Ans	02	M	LA
59	Merchant Name	Ans	Var up to 25	M	Must provide every time

60	Merchant city	Ans	Var up to 15	M	Vientiane
63	Cyclic Redundancy Check		04	M	XXXX CRC implement

18.1.2 Dynamic QRC

- Tag ID 52, the Merchant Category Code can be used by the acquirer or UnionPay if they use it themselves. The acquirer needs to convert/match the code to the UnionPay code at the system backend for transaction processing. Please refer to the file titled "Updated UPI MCC List (Including Newly-added MCCs).pdf" for the MCC information.
- Tag ID 53, the Transaction Currency will only use the LAK code, which is 418.
- Tag ID 54, the Transaction Amount tag should be mandatory for dynamic QRC, as it is generated by the merchant.
- Tag 59 is the Merchant Name, which is mandatory for UnionPay. Please raise this as a requirement to add.
- Tag ID 62, the Additional Data Field Template, it is suggested to follow the definition of EMVco.

Example for Dynamic QRC

ID	Data Object	Type	Length	Presence	Description or Example
00	Payload format Indicator	N	02	M	01
01	Point of Initiation method	N	02	O	12
15	UnionPay Tag	Ans	Var up to 35	M	Acquirer IIN + Forwarding IIN + Merchant ID
	Acquirer IIN	Ans	Var up to 8	M	00010344
	Forwarding IIN	Ans	Var up to 8	M	00010344
	Merchant ID	Ans	Var up to 15	M	ABCDEFGHIJKLMNO
38	Lao National QRC for QR payment	Ans			
00	AID	Ans	16	M	A005266284662577
01	<u>IIN</u>	N	08	M	27710418 detail
02	<u>Payment type</u>	N	03	M	002 detail
03	Receiver ID	Ans	Var up to 25	M	ABCDE123456789
52	<u>Merchant Category code</u>	N	04	M	5813
53	Transaction Currency	Ans	03	M	418 (LAK)
54	Transaction Amount	N	Var up to 13	C	250000
58	Country Code	Ans	02	M	LA
59	Merchant Name	Ans	Var up to 25	M	Must provide every time
60	Merchant city	Ans	Var up to 15	M	Vientiane
62	Additional Data Field Template	Ans	Var up to 99	O	
01	Bill Number	Ans	Var up to 25	O	00000100
02	Mobile number	Ans	Var up to 25	O	7XXXXXXXX
03	Store Label	Ans	Var up to 25	O	SYHOME
04	Loyalty number	Ans	Var up to 25	O	

05	Reference Label	Ans	Var up to 25	O	00000021
06	Customer Label	Ans	Var up to 25	O	
07	Terminal Label	Ans	Var up to 25	O	T00001
08	Purpose of Transaction	Ans	Var up to 25	O	Iced coffee
09	Additional Consumer Data Request	Ans	Var up to 3	O	
63	Cyclic Redundancy Check		04	M	XXXX CRC implement

18.2 Incoming Inquiry and Transfer (UnionPay-Lao)

When requested by UnionPay "frommember": "CHINA_UPI" is must be Follow up process domestic and **it must only be completed**

19 Cambodia Cross border Payment

19.1 Incoming Inquiry and Transfer (Cambodia-Lao)

When requested by Cambodia is must be Follow up process domestic

19.2 Outgoing Inquiry (Lao-Cambodia)

19.2.1 Request to send to LAPNET

19.2.1.1 URL

<http://<LAPNET-IP>:8055/inquiry>

19.2.1.2 Request Headers

Content-Type: application/json

X-Source-Request-Signature: <<source Request signature>>

19.2.1.3 Request Body

Field	Type	Max-Length	Description
frommember	String	16	E.g., BCEL, LDB, APB, LVB
fromuser	String	64	E.g., adsavin@gmail.com
fromaccount	String	32	E.g., 010120000123456001
totype	String	16	QR
toaccount	String	2048	Account number or card number or full QR data
tomember	String	16	Only required if totype is ACCOUNT
reference	String	64	Member-generated unique reference
time	String	24	e.g.: 2023-10-09 05:34:11

19.2.1.4 Example of request body

```
{
  "frommember": "BCEL",
  "fromuser": "DJ3LD0217BPC830HDHZ8RZRFA",
  "fromaccount": "DJ3LD0217BPC830HDHZ8RZRFA",
  "totype": "QR",
  "toaccount":
    "00020101021130360009khqr@ac1b01090126699410206ACLEDA3920001185512463062010
    115204701153031165802KH5922P WALLET NOT LINK T246010Phnom
    Penh621302090126699416304F987",
  "tomember": "CAMBODIA",
  "reference": "BONE1408KKDWXNYR",
  "time": "2023-10-13 08:49:03"
}
```

19.2.1.5 Response coming from LAPNET

19.2.1.6 Response Headers

Content-Type: application/json
 X-Source-Request-Signature: <<source request signature>>
 X-LAPNET-Request-Signature: <<LAPNET request signature>>
 X-Destination-Response-Signature: <<destination response signature>>
 X-LAPNET-Response-Signature: <<LAPNET response signature>>
 X-LAPNET-Transaction-ID: <<Sequence transaction ID generate by LAPNET>>

19.2.1.7 Response Body

Field	Type	Max-Length	Description
result	String	64	See Result Table
reference	String	64	Original request reference
time	String	25	Response time
originalmessage	JSON object		Original request message
accountname	String	128	Account name, card name, merchant name
accountlocation	String (Optional)	256	
accountphone	String (Optional)	24	
invoice	String (Optional)	128	
terminal	String (Optional)	128	
amount	Number (24,2)		If transaction has fixed amount, set it here
purposerequired	Boolean (Optional)	–	- For type ACCOUNT and CARD, the default value is true. - For type QR, default value is false
purpose	String (Optional)	512	If transaction has a fixed purpose, set it here
accountccy	String separated by comma	16	E.g. "LAK", or "LAK,USD"

feelist	JSON Object		Fee list calculated and injected by LAPNET
crossborderexchangerate	JSON Object		

19.2.1.8 Example Response Body

```
{
  "result": "OK",
  "accountname": "P WALLET NOT LINK T24",
  "accountlocation": "Phnom Penh",
  "accountphone": "012669941",
  "accountccy": "KHR",
  "bank": "CAMBODIA",
  "originalmessage": {
    "frommember": "BCEL",
    "fromuser": "DJ3LDO217BPC830HDHZ8RZFA",
    "fromaccount": "DJ3LDO217BPC830HDHZ8RZFA",
    "totype": "QR",
    "toaccount":
      "00020101021130360009khqr@ac1b01090126699410206ACLEDA392000118551246306201
      0115204701153031165802KH5922P WALLET NOT LINK T246010Phnom
      Penh621302090126699416304F987",
    "tomember": "CAMBODIA",
    "reference": "BONE1408KKDWNXYR",
    "time": "2023-10-13 08:49:03"
  },
  "feelist": {
    "LAK": [
      {
        "from": 0,
        "feeamount": 0
      },
      {
        "from": 999999999,
        "feeamount": 0
      }
    ]
  },
  "crossborderexchangerate": {
    "id": 3,
    "ccy": "KHR",
    "lastupdate": "2023-09-26 22:45:47",
    "rate": "5.20"
  }
}
```

19.3 Outgoing Transfer (Lao-Cambodia)

19.3.1 Request to send to LAPNET

19.3.1.1 URL

<http://<LAPNET-IP>:8055/transfer>

19.3.1.2 Request Headers

Content-Type: application/json

X-Source-Request-Signature: <<source Request signature>>

19.3.1.3 Request Body

Field	Type	Max-Length	Description
frommember	String	16	E.g., BCEL, LDB, APB, LVB
fromuser	String	64	E.g., narin@gmail.com
exchangerate	String	26	E.g., "5.2" (must be equal to crossborderexchangerate , refer to 2.2.2 inquiry response)
fromaccount	String	32	E.g., 010120000123456001
totype	String	16	ACCOUNT/CARD/QR
toaccount	String	2048	Account number or card number or full QR data
tomember	String	16	Only required if totype is ACCOUNT
reference	String	64	Member-generated unique reference. Should be the same as inquiry reference.
time	String	25	e.g.: 2020-10-09 05:34:11
amount	Number (24,2)		If the transaction has a fixed amount, this amount must be the same as the amount received when inquired.
fee	Number (24,2)		Fee calculated by member according to "feelist". LAPNET will double-check this, if the fee not matched, LAPNET will reject with result FEE_NOT_CORRECT
ccy	String	3	Settlement CCY. If sender specifies USD, sending bank will need to settle in USD
purpose	String	512	If totype is ACCOUNT, the purpose is required. Purpose's requirement can be overridden by field purpose required when inquired.

19.3.1.4 Example

```
{
  "frommember": "BCEL",
  "fromuser": "DJ3LD0217BPC830HDHZ8RZRFA",
  "fromaccount": "0100120000000004770001",
  "totype": "QR",
```

```

    "toaccount":
    "00020101021130410009khqr@ac1b0114000103535631020206ACLEDA3920001185512466247010
    115204593153031165802KH5925VN SCANS KHQR BY COUNTER16010Phnom
    Penh622702090934439690710000000025663046C1B",
    "tomember": "CAMBODIA",
    "reference": "BONELBKRD6YA45G1",
    "time": "2023-10-13 08:53:25",
    "amount": 200000,
    "fee": 0,
    "exchangerate": "5.20",
    "ccy": "LAK",
    "purpose": "PAYMENT"
}

```

19.3.2 Response coming from LAPNET

19.3.2.1 Headers

```

Content-Type: application/json
X-Source-Request-Signature: <<source request signature>>
X-LAPNET-Request-Signature: <<LAPNET request signature>>
X-Destination-Response-Signature: <<destination response signature>>
X-LAPNET-Response-Signature: <<LAPNET response signature>>
X-LAPNET-Transaction-ID: <<Sequence transaction ID generate by LAPNET>>

```

19.3.2.2 Body

Field	Type	Max-Length	Description
result	String	64	See Result Table
reference	String	64	Original request reference
shouldrevert	Integer 0 or 1		When the result is not "OK", if "shouldrevert" is 1 then refund the customer. Otherwise, follow the procedure when transfer fails. Use Integer instead of Boolean to make database record easier
time	String	25	e.g.: 2020-10-09 05:34:11
originalmessage	JSON object		Original request message

19.3.2.3 Example

```

{
  "result": "OK",
  "reference": "BONE3P19031GOC2X",
  "time": "2023-10-13 08:49:26",
  "originalmessage": {
  },
  "shouldrevert": 0
}

```

